WELCOME TO SI-WATER

Let’s get your tank online.

This guide walks you through unpacking, mounting and connecting your SI-Water smart tank monitor to Home Assistant. Most installs take about 15 minutes. If you get stuck at any step, we’re an email away at **sales@solairesa.co.za**.

WHAT’S IN THE BOX

Check your contents

- SI-Water IP65 controller (pre-wired & tested)
- Stainless-steel submersible probe with ~5 m cable
- 220 V AC mains lead with plug
- Cable gland (fitted)
- This Quick Start Guide

WHAT YOU’LL NEED

Have these ready

- A drill & screws (for wall-mounting the IP65 box)
- A tape measure (to measure tank height when full)
- A smartphone or laptop (for Wi-Fi setup)
- Your Home Assistant dashboard

AT A GLANCE

The 8 steps ahead

1	Mount the IP65 box
2	Fit the probe in the tank
3	Power it on
4	Connect to the SI-Water Wi-Fi
5	Join it to your home Wi-Fi
6	Add it to Home Assistant
7	Calibrate — tank height, volume & low-level
8	Add to your dashboard

1 Mount the IP65 box

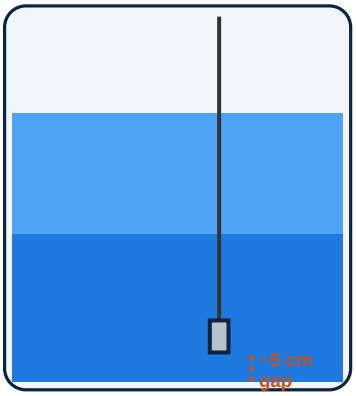
Choose a spot on the side of your tank (or on the wall next to it) where the controller will stay dry and get good Wi-Fi. Wall-mount the IP65 enclosure using the four fixing points so that the cable gland points **downwards**.

- Keep the box above the **highest possible** water level in the tank.
- Leave ~15 cm clearance around the box for cable access.
- Avoid direct sun on the enclosure where possible.

> **■ Important — mount the box ABOVE the water line**
>
> The probe uses an internal air tube to reference atmospheric pressure. If the controller box ends up **below** the probe and the tank overflows or back-fills, water can be siphoned up the cable through that air tube and damage the electronics. **Always mount the box above the highest possible water level.**

2 Fit the probe in the tank

Feed the probe on its cable through a sealed cable entry on the tank lid (or a pre-drilled & sealed hole). Lower the probe until it hangs **just above** the bottom of the tank — about 5 cm clear.



Why 5 cm off the bottom?

Silt, sand and debris collect at the very bottom of every tank. Hanging the probe a few centimetres above that layer keeps the pressure port clear and the readings accurate. You'll lose a few litres off the bottom of the reading — you can account for this in calibration (Step 7).

Tie the cable off cleanly at the top so the probe can't swing against the tank wall. Don't kink the cable — it carries the air tube that makes the sensor work.

> **■ Pro tip — don't pinch the cable**
>
> The probe cable contains a tiny air tube that lets the sensor reference atmospheric pressure. A crushed or pinched cable will give inaccurate readings. Use a cable gland or grommet rather than a tight clamp.

3 Power it on

Plug the SI-Water mains lead into a normal 220 V socket. The controller will boot up in a few seconds. There's a small status LED on the board — visible through the cover — that tells you at a glance what the unit is doing.

	Colour	Means	What to do
🟡	**Yellow**	Waiting for pairing	The unit has booted but isn't yet on a Wi-Fi network. Follow Step 4.
🔵	**Blue**	Wi-Fi connected	SI-Water is on your home Wi-Fi and talking to Home Assistant.
🟢	**Green**	Signal received from probe	All good — pressure readings are flowing. This is the normal running state.
🔴	**Red**	Signal lost or no signal	The controller isn't hearing the probe. Check the cable at the gland and at the board.

A healthy SI-Water will pass through **yellow** → **blue** → **green** within about a minute of first power-on once you've completed the Wi-Fi steps below.

4 Connect your phone to the SI-Water hotspot

On first boot, SI-Water puts up its own temporary Wi-Fi hotspot so you can tell it which home Wi-Fi to join. On your phone (or laptop):

Network (SSID)	`Si Water`
Password	`12345678`
Band	2.4 GHz only

Most phones will automatically open a **Captive Portal** page as soon as you connect — the SI-Water setup screen appears by itself. If it doesn't, open a browser and go to **http://192.168.4.1**.

5 Join your home Wi-Fi

On the captive portal screen, pick your home Wi-Fi network from the list and enter its password.

- Pick the **same Wi-Fi network** your Home Assistant server is on.
- **2.4 GHz only.** If your router uses one combined SSID, that's fine — SI-Water will pick the 2.4 GHz band.
- Hit **Save**. SI-Water restarts and joins your network (about 30 seconds).

■ Weak Wi-Fi at the tank?

Tanks are usually outside, which is often where home Wi-Fi is weakest. A small Wi-Fi extender or mesh node near the tank solves 90% of connection problems. Ask us if you need a recommendation.

6 Add it to Home Assistant

Once SI-Water is on your home network, Home Assistant will discover it automatically via ESPHome. You'll see a notification:

Settings → **Devices & Services** → **Discovered** — an entry named **"Water Tank Sensor"** (or similar) will appear.

Click **CONFIGURE**, accept the ESPHome integration, and SI-Water's entities (Tank Level %, Tank Volume L, Water Flow Rate, etc.) will be added to Home Assistant.

■ Not running Home Assistant yet?

SI-Water needs Home Assistant to show you the data. Home Assistant is free, runs on a Raspberry Pi or any small computer, and the setup takes about an hour. We're happy to walk you through it — or put you in touch with an approved installer. Email **sales@solairesa.co.za**.

7 Calibrate it to your tank

SI-Water ships with safe defaults, but you'll get accurate litres and percentage by telling it the real dimensions of your tank. In Home Assistant, go to the **Water Tank Sensor** device page:

Setting	What to enter
Tank Height (cm)	Measure (or look up) the internal height of your tank from the bottom to the maximum water mark. Set this value.
Tank Total Volume (L)	Enter the tank's nominal full volume in litres (e.g. 2,500 / 5,000 / 10,000). Check the moulding — most tanks have this printed on the side.
Low Level Warning (%)	Set the percentage at which you want to be notified. 20% is a sensible default — lower if your tank refills automatically.
Reset Water Totals	Tap once after install to zero the daily/monthly counters cleanly.

8 Add SI-Water to your dashboard

In Home Assistant, pick any Lovelace dashboard and use **Add Card** to drop in the SI-Water entities. Common starter cards:

- **Gauge** card on Tank Level (%) — at-a-glance view, with amber/red thresholds.
- **Entity** card on Tank Volume (L) — exact litres in the tank right now.
- **History** graph on Tank Volume (L) — see fills and usage over the last 24 h / week.
- **Entities** card with Water Flow Rate, Daily/Monthly fill & usage for detail.
- **Conditional** card to show a red warning when Low Tank Warning is ON.

> **Want a ready-made dashboard?**
> Email us and we'll send you a Lovelace YAML snippet you can paste in for an instant SI-Water overview tile.

TROUBLESHOOTING

Start with the LED

The status LED is your fastest diagnostic. Read its colour first, then use the table below to narrow down the problem.

	Colour	What it means	Try this
	Yellow	Waiting for pairing — unit is on but not on Wi-Fi	Re-do Steps 4 & 5: join the **Si Water** hotspot and enter your home Wi-Fi credentials in the captive portal. If the hotspot doesn't appear, power-cycle for 10 s and try again.
	Blue	On Wi-Fi, but no probe data yet	Wi-Fi is fine. Give it 30–60 s to establish the probe link. If it stays blue, check the probe cable termination inside the IP65 box — a loose screw terminal is the usual culprit.
	Green	All good — pressure data flowing	Normal running state. If readings look odd, jump to the Calibration & Readings rows below.
	Red	Signal lost or no signal from probe	The controller isn't hearing the probe. Check (1) the probe cable is seated in its terminal block, (2) the cable gland is tight but not crushing the cable, (3) the cable hasn't been kinked (this pinches the internal air tube).
	Off / dark	No power to the controller	Check the 220 V mains lead, the plug, and the wall socket. If mains is present and the LED is still dark, contact us — don't open the enclosure while live.

Other common questions

Symptom	Try this
SI-Water hotspot doesn't appear	LED should be **yellow**. If it isn't, power-cycle for 10 s. Then check your phone's Wi-Fi list for **Si Water**. Some phones hide 'open' networks — toggle Wi-Fi off/on.
Can't see it in Home Assistant	LED should be **blue** or **green**. Confirm your phone and HA server are on the **same** Wi-Fi network. Restart Home Assistant. It should appear under Settings → Devices & Services → Discovered.
Readings look wrong at full / empty	Re-check Tank Height and Tank Total Volume. Remember the probe sits ~5 cm off the bottom; the last few litres won't register. You can offset in calibration.
Readings drift or are unstable	Usually a pinched cable or moisture at the cable gland. Inspect the gland first. If the LED flickers between green and red, the cable connection is intermittent.
Flow readings jump around	Brief spikes are normal (ball valve dribble, thermal expansion). Use daily totals rather than instantaneous flow for billing or usage reporting.
Low-level alert won't trigger	Make sure the Low Level Warning slider is set above the current tank %. Also check your HA notification / automation is enabled.
Wi-Fi drops out intermittently	Tank position is usually the weakest Wi-Fi spot in the yard. LED will cycle yellow ↔ blue. A Wi-Fi extender or mesh node near the tank solves this in nearly every case.

Need a hand?

We're here to help — whether it's a stuck calibration, a Wi-Fi issue, or a full Home Assistant setup from scratch.

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